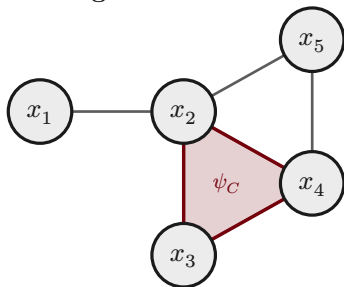


Markov random field — undirected graphical model

general MRF



maximal clique $C = \{x_2, x_3, x_4\}$

$$p(x) = \frac{1}{Z} \prod_C \psi_C(x_C)$$

$$Z = \sum_x \prod_C \psi_C(x_C) \text{ (partition function)}$$



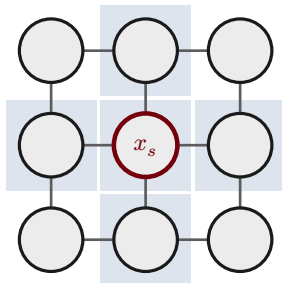
variable x_i

— undirected edge (no direction)



maximal clique (potential ψ_C)

grid (lattice) MRF



Markov blanket of x_s = its 4 neighbours

$$x_s \perp x_{\text{rest}} \mid x_{N(s)}$$